

Data Structures & Algorithms: Design, Proof & Performance

Models, formulas & decision rules. Read the full notes for derivations and worked examples.

01 / Name the guarantee

Worst-case maximizes over inputs; expected averages over stated randomness; amortized averages over a sequence. Input size and arithmetic cost model must be explicit before interpreting any bound.

02 / Representation tradeoffs

Array indexing is $O(1)$, interior shifts $O(n)$. Doubly linked edits at known positions are $O(1)$, but locating an index is $O(n)$. Include both search and modification costs.

03 / BST versus AVL

BST search and update cost $O(h)$; range reporting costs $O(h+k)$. AVL maintains child-height difference at most one, ensuring $h=O(\log n)$. k is output size.

04 / Heap priorities

A min-heap keeps each parent no larger than its children. Minimum is $O(1)$; insert and remove-min are $O(\log n)$. A heap array is not fully sorted.

05 / Hash-table assumptions

With suitable uniform hashing, chaining lookup is expected $O(1+\alpha)$, $\alpha=n/m$. Linear probing needs tombstones or cluster repair after deletion. Cuckoo insertion can trigger expensive rebuilding.

06 / Graph traversals

Adjacency-list BFS and DFS cost $O(n+m)$. BFS minimizes edge count. Undirected DFS tree edge (u,v) is a bridge when $\text{low}[v] > \text{disc}[u]$, with parent edges handled correctly.

07 / Path versus tree

Dijkstra relaxes $d[u] + w(u,v)$ and requires nonnegative weights. Prim uses $w(u,v)$ to connect a new vertex. MSTs minimize total network weight, not source-to-vertex distances.

08 / Greedy proof certificates

Fractional knapsack uses density and equal-weight exchange. Interval partitioning matches overlap depth. Huffman combines two minimum frequencies. Each proof depends on its specific feasible-choice structure.

09 / Recurrences

$T(n) = T(n/2) + \Theta(1)$ gives $\Theta(\log n)$; $2T(n/2) + \Theta(n)$ gives $\Theta(n \log n)$; $3T(n/2) + \Theta(n)$ gives $\Theta(n^{\log_2 3})$. Include split and combination costs.

10 / Sort versus select

Randomized quicksort sorts both sides in expected $O(n \log n)$. Quickselect follows one side in expected $O(n)$. Both can take $\Theta(n^2)$ on unlucky pivot sequences.

11 / Random structures

Fisher-Yates chooses j uniformly in $\{0, \dots, i\}$ before fixing position i . Skip-list promotion probability one half gives expected $O(\log n)$ operations and expected $O(n)$ storage.

12 / Read the input length

For q encoded in $k = \log_2 q$ bits, $\sqrt{q} \approx 2^{(k/2)}$. A polynomial in the numerical value may be exponential in bit length; this distinction matters for cryptographic complexity.